# Saakash Sathyan

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## Education

**University of California, Irvine**

*B.S. Aerospace Engineering* — Expected June 2027

- GPA: 3.25
- Relevant Coursework: Dynamics, Fluid Dynamics I-III, Thermodynamics, Astronautics, Circuits, Materials Science, Aerodynamics, Mechanics of Structures, Mechanical Systems Lab, Introduction to Control Systems, Dynamics and Control of Aerospace Vehicles, Propulsion, Aircraft Performance

## Projects

**AI-Integrated Companion Robot** | *Project Lead* — Jul 2026 – Present

- Collaborating with an industry mentor (System Architect, Google) and two peers to design an AI-driven companion robot
- Developed initial concept sketches and block-component CAD models defining system layout and basic dimensioning
- Built bills of materials with component trade-offs across compute, sensing, and actuation subsystems
- Researching split compute architecture utilizing Raspberry Pi for AI processing and ESP32 for motion control
- Evaluating sensor and camera configurations for obstacle avoidance, edge detection, and voice interaction

**UCI MAE 106 Autonomous Pneumatic Robot** | *Mechanical Coordinator* — Mar 2026 – Jun 2026

- Designed full chassis assembly in SolidWorks including frame, wheel mounts, and steering linkage
- Engineered pneumatic propulsion system harnessing a single cylinder powered by a compressed air tire
- Designed steering mechanism actuated by servo motor to enable autonomous directional control
- Fabricated components using band saw, drill press, laser cutter, and 3D printer with PLA filament
- Robot competed autonomously using bearing-based orientation and wheel-rotation odometry for navigation

**SEDS @ UCI Autonomous Mars Rover** | *Fabrication Subteam* — Nov 2025 – Jun 2026

- Designed autonomous rover to traverse uneven terrain and retrieve a rock sample without human input
- Modeled full chassis assembly in SolidWorks including dual-deck frame, 4 motor mounts, and wheel hubs
- Integrated 3 subassemblies (turning mechanism, 2-DOF arm, claw) into the full rover assembly
- Created system-level concept sketches defining rover architecture and component relationships across 3 subteam
- Fabricated arm, claw, and turning mechanism components using Orca Slicer and Bambu X1C 3D printers

## Leadership & Activities

**Participant** | **Project Manager** — Sep 2026 – Present

*NASA L'SPACE Mission Concept Academy* — *Remote*

- Lead a 21-member multidisciplinary student team through NASA's L'SPACE Mission Concept Academy program
- Coordinate task delegation, scheduling, and milestone tracking across engineering science, and programmatic subteams
- Develop mission documentation and technical deliverables following NASA systems engineering standards
- Utilize Siemens NX for mission hardware modeling as part of the concept development process

**Mentor** — Sep 2026 – Present

*Aero Academy | AIAA @ UCI* — *Irvine, CA*

- Serve as a founding mentor in AIAA @ UCI's new mentorship program connecting board members with incoming students
- Guide a group of 4 mentees through hands-on aerospace projects, beginning with individual foam glider builds
- Instruct mentees on fundamental aerodynamics and flight physics as applied to fixed-wing glider design
- Facilitate social activities to foster community and belonging within the chapter
- Prepare mentees for competitive project teams such as Design Build Fly through foundational project experience

**Board Member | Internal Affairs Director, Financial Director** — Nov 2025 – Present

*AIAA @ UCI* — *Irvine, CA*

- Organize and facilitate general chapter meetings and manage on-campus event logistics
- Lead recruitment efforts and coordinate member engagement across a 100+ member organization
- Oversee budget planning and financial reporting for a ~$10,000 annual budget
- Personally managed 2 fundraising events from planning through execution and financial reconciliation
- Coordinated sponsorship acquisition and developed proposals for aerospace industry partners

## Technical Skills

**Programming**: MATLAB, Python, C++
**CAD & Design**: SolidWorks, Siemens NX, OnShape, ANSYS, Orca Slicer, Bambu Studio
**Engineering**: Mechanical Design, CAD Modeling, System Integration, Aerodynamic Analysis, CFD Simulation, Technical Documentation, Design Review, GD&T
**Fabrication & Hardware**: Hand & Power Tools, Band Saw, Belt Sander, FDM 3D Printing, Soldering, Oscilloscope, Function Generator, Digital Calipers
**Productivity**: Microsoft Office, Google Suite